Stochastic Differential Equations – Notes written by Ike Glassbrook, based on the course lectured by Massimiliano Gubinelli in MT 2025, and the lecture notes of Zhongmin Qian, 2021

Stochastic calculus revision

We begin by recalling some of the key results which allow us to work well with semimartingales.

Theorem 1 *Let (X_t) be a super-martingale on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$, $D \subseteq [0, \infty)$ countable:*

- *The process of right limits $Z_t = \lim_{\substack{s\in D\\ s\downarrow t}} X_s$ of (X_t) in D is well-defined a.s., is $\mathcal{F}_t^+$-measurable, and has left limits a.s..*
- *$\mathbb{E}\big[Z_t \,|\, \mathcal{F}_t\big] \le X_t$ a.s..*
- *(Z_t) is a super-martingale a.s. on $(\Omega, \mathcal{F}, (\mathcal{F}_t^+), \mathbb{P})$.*

Note that (Z_t) can still differ somewhat from (X_t) if (X_t) is not a martingale (and in particular, if $t \mapsto \mathbb{E}\big[X_t\big]$ is not right-continuous), as we possibly don't have that (Z_t) is a modification of (X_t).

Definition 1 (Usual conditions) *A filtration $(\mathcal{F}_t)$ satisfies the usual conditions if $(\Omega, \mathcal{F}, \mathbb{P})$ is a complete probability space, $(\mathcal{F}_t)$ is right-continuous, and $\mathbb{P}^{-1}(\{0\}) \subseteq \mathcal{F}_t$ for all $t \ge 0$.*

Definition 2 (Local martingales) *(M_t) is a local martingale if it is right-continuous, and there is a sequence (T_n) of stopping times such that $T_n \to \infty$ as $n \to \infty$ a.s., and $(M_{t\wedge T_n})$ is a martingale for all $n \ge 1$ (with respect to a fixed filtration $(\mathcal{F}_t)$).*

Right-continuity makes our lives easier by ensuring that we can talk about hitting times.

Our intuition should be that a continuous martingale is just a continuous local martingale plus an integrability condition (does not blow up too quickly). A continuous local martingale localises in such a way as to 'catch' each sample trajectory at a time when it is known to be of reasonable size, thus giving us a martingale by muting the blow-up of unfavourable trajectories.

To understand more, consider a non-negative random variable of infinite expectation, ξ. Defining $X_t := B_{\xi t}$, we have for each ξ a process which looks locally like a martingale, but is not integrable due to large tails of ξ giving rise to arbitrarily quick growth trajectories. In order to understand this process, we therefore want to catch each trajectory once it reaches a certain size, e.g. hits $\{\pm n\}$, and at this point we've muted the behaviour of high-ξ trajectories while keeping the reasonable size ones, so as to preserve integrability while getting a slightly weaker martingale property.

We recall the following results

1. A non-negative continuous local martingale such that $M_0 \in L^1$ is a supermartingale.
2. A continuous local martingale dominated from above by an integrable random variable for $t \ge 0$ is a UI martingale.
3. For any continuous local martingale such that $M_0 \in L^1$, the hitting times of $\mathbb{R} \setminus (-n, n)$ are a reducing sequence.
4. For any continuous local martingale, stopping it at a stopping time gives us another continuous local martingale.

Note that (3) gives us that bounded continuous local martingales are local martingales.

Proposition 1 *For a martingale (M_t), $s < t$,*

$$\mathbb{E}\left[M_t^2 - M_s^2 \,\middle|\, \mathcal{F}_s\right] = \mathbb{E}\left[(M_t - M_s)^2 \,\middle|\, \mathcal{F}_s\right].$$

This is a matter of algebra, just observing that $\mathbb{E}\left[M_t M_s - M_s^2 \,\middle|\, \mathcal{F}_s\right] = 0$. With this, it follows that any continuous local martingale is of finite variation iff it is a.s. identically zero (for the square expectation, stopped at a hitting time of the finite variation, is the quadratic variation, which must be zero if the continuous local martingale is of finite variation).

Definition 3 (Quadratic variation) *For $M \in \mathcal{H}^{2,c}$,*

$$\langle M\rangle_t := \lim_{\mathrm{mesh}(t_i)\to 0} \sum_{i=1}^{|\pi|} |M_{t_i} - M_{t_{i-1}}|^2,$$

where the limit is taken in probability and L^2, is the quadratic variation of M.

This definition may then be extended to continuous local martingales by a theorem of Le Gall:
Theorem 2 *For M a continuous local martingale. There exists a a.s. unique non-decreasing, continuous adapted finite variation process $(\langle M\rangle_t)$ started at zero such that $(M_t^2 - \langle M\rangle_t)$ is a continuous local martingale. For any $t > 0$, and sequence of partitions $(t_i^{(n)})$ with mesh tending to zero, in probability we have the limit*

$$\langle M\rangle_t = \lim_{n\to\infty} \sum_{i=1}^{|(t_k^{(n)})|} (M_{t_i^{(n)}} - M_{t_{i-1}^{(n)}})^2.$$

It may then be extended to semimartingales, noting that the quadratic variation of any finite variation process is identically zero.

Theorem 3 *Let M be a continuous local martingale with $M_0 \in L^2$.*

In the weak case, M is a martingale which is square integrable for all $t \ge 0$ iff $\mathbb{E}\left[\langle M\rangle_t\right] < \infty$ for all $t \ge 0$. Furthermore, in this case $M_t^2 - \langle M\rangle_t$ is a martingale.

In the strong case, M is a martingale such that $\sup \mathbb{E}\left[M_t^2\right] < \infty$ iff $\mathbb{E}\left[\langle M\rangle_\infty\right] < \infty$, and in this case $M_t^2 - \langle M\rangle_t$ is a UI martingale.

The definition and results concerning the covariation are then entirely analogous. Note in particular that we can achieve these results by polarisation, that

$$\langle M, N\rangle := \frac{1}{2}\left(\langle M+N\rangle - \langle M\rangle - \langle N\rangle\right).$$

Furthermore, recall Kunita-Watanabe, that

$$\int_0^t |H_s||K_s||\mathrm{d}\langle M, N\rangle_s| \le \left(\int_0^t H_s^2 \mathrm{d}\langle M\rangle_s\right)^{1/2} \left(\int_0^t K_s^2 \mathrm{d}\langle N\rangle_s\right)^{1/2}.$$

From here, we now just recall that integration against continuous martingales M is defined for those progressively measurable processes K such that for $t \ge 0$,

$$\int_0^t K_s^2 \mathrm{d}\langle M\rangle_s < \infty.$$

While integrating against a $\mathcal{H}^{2,c}$ martingale gives a $\mathcal{H}^{2,c}$ martingale, this integration gives another continuous martingale. We then obtain the commutativity relation

$$\left\langle \int_0^\cdot K_s \mathrm{d}M_s, N\right\rangle_t = \int_0^t K_s \mathrm{d}\langle M, N\rangle_s$$

and the associativity relation

$$\int_0^t K_s H_s \mathrm{d}M_s = \int_0^t K_s \mathrm{d}\left(\int_0^s H_u \mathrm{d}M_u\right).$$

Finally, before our key result, we note an integration by parts formula, for continuous semimartingales:

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s \mathrm{d}Y_s + \int_0^t Y_s \mathrm{d}X_s + \langle X, Y\rangle_t.$$

This may be proved using the Riemann integral limit formulation of stochastic integration, taking a sum over a partition and then taking the mesh to zero. This limit is only in probability, but it is sufficient to note uniqueness in this case.

Theorem 4 (Itô's formula) *For $X = (X_t^{(i)})_{i\in[d]}$ a continuous semimartingale in $\mathbb{R}^d$ with decomposition $X = M + A$ where $M^{(i)}$ are continuous local martingales, and $A^{(i)}$ are continuous locally integrable adapted processes with finite variations, let $f \in C^2(\mathbb{R}^d, \mathbb{R})$. Then*

$$f(X_t) - f(X_0) = \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x_i}(X_s)\,\mathrm{d}X_s^{(i)} + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j}(X_s)\,\mathrm{d}\left\langle M^{(i)}, M^{(j)}\right\rangle_s.$$

This is often written equivalently as

$$\mathrm{d}f(X_t) = \sum_{i=1}^d \frac{\partial f}{\partial x_i}(X_s)\mathrm{d}X_t^{(i)} + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d \frac{\partial^2 f}{\partial x_i \partial x_j}\,\mathrm{d}\left\langle M^{(i)}, M^{(j)}\right\rangle_t.$$

The proof of this formula is derived almost entirely via the Stone-Weierstrass theorem stating that polynomials on compact sets $K \subseteq \mathbb{R}^n$ are dense in $C(K)$. Consequently, it should be emphasised that, contrary to how the proof is often stated, we do not require that $f \in C^2(\mathbb{R})$, but merely $C^2(K)$ where K is of sufficient size to be well-defined in the context of the formula.

Theorem 5 (Levy's characterisation) *Let $M = (M_t^{(i)})_{i\in[d]}$ be an adapted continuous stochastic process with initial value 0. Then M is a Brownian motion iff each coordinate is a square integrable martingale, and $\left\langle M^{(i)}, M^{(j)}\right\rangle_t = \delta_{ij} t$ for each pair.*

This is seen by a generating function argument using Itô's formula on $x \mapsto e^x$ with the appropriate semimartingale.

Theorem 6 (Dambis-Dubins-Schwarz) *Let M be a continuous local martingale starting at 0, where $\langle M\rangle_\infty = \infty$. Then with T_t the hitting time of t by $\langle M\rangle$, M_{T_t} is a $\mathcal{F}_{T_t}$ Brownian motion, and $M_t = B_{\langle M\rangle_t}$.*

While the $\langle M\rangle_\infty = \infty$ condition seems problematic for the application of this theorem, it turns out to be of no matter at all when working on a finite interval: we can just glue a Brownian motion to run starting at M_T for $t > T$, and this gives us a time-changed Brownian motion for $[0, T]$ (so running on $[0, \langle M\rangle_T]$ in the original BM).

Further results in stochastic calculus

Theorem 7 (BDG inequality) *Let X be a continuous local martingale starting at 0, $T > 0$:*

- *for $p > 1$*

$$(4p)^{-p}\mathbb{E}\left[\langle X\rangle_T^p\right] \le \mathbb{E}\left[|\sup_{t\le T} X_t|^{2p}\right] \le (2ep^2)^p\mathbb{E}\left[\langle X\rangle_T^p\right]$$

- *for $p = 1$*

$$\mathbb{E}\left[\langle X\rangle_T\right] \le \mathbb{E}\left[|\sup_{t\le T} X_t|^2\right] \le 4\mathbb{E}\left[\langle X\rangle_T\right]$$

- *For $0 < p < 1$*

$$p^{2p}\mathbb{E}\left[\langle X\rangle_T^p\right] \le \mathbb{E}\left[|\sup_{t\le T} X_t|^{2p}\right] \le \left(\frac{16}{p}\right)^p \mathbb{E}\left[\langle X\rangle_T^p\right]$$

Read these inequalities as encompassing that $(X^*)^{2p}$ and $\langle X\rangle^p$ are of the same mean asymptotic growth. Their proofs are a combination of Doob's maximal inequality for submartingales, Itô's formula applied with appropriate functions, IBP, and Hölder's inequality. Note that these inequalities can all be attacked with MCT, as all of the random variables concerned are increasing in T, and this means we can straightforwardly localise back to the true martingale case.

Exponentials

Recall from CMSC that we may define for any continuous semimartingale,

$$\mathcal{E}(X)_t := \exp\left(X_t - \frac{1}{2}\langle X\rangle_t\right)$$

as the stochastic exponential of X, and the unique solution to

$$\mathrm{d}Z_t = Z_t \mathrm{d}X_t$$

with $Z_0 = 1$. That this is a solution may be seen via Itô's formula (and the derivation via an ansatz). Uniqueness is shown later **add uniqueness**

The special case of this is of course when $X_t = t$, and we recover e^x.

If X is just a continuous local martingale, immediately $\mathcal{E}(X)$ is, as a non-negative continuous local martingale, a supermartingale. In fact we can go further, by arguing that under regularity conditions, $\mathcal{E}(\alpha M)$ is UI for $\alpha \in (0, 1)$, and thus taking $\alpha \to 1$ we see that $\mathbb{E}\left[\mathcal{E}(M)\right] = 1$ so it is a martingale.

Theorem 8 (Novikov) *For M a continuous local martingale started at 0, if*

$$\mathbb{E}\left[\exp\left(\frac{1}{2}\langle M\rangle_T\right)\right] < \infty$$

then $(\mathcal{E}(M)_{t\wedge T})$ is a martingale.

Unfortunately, this condition is often quite difficult to verify, and thus we more often will retreat to the supermartingale characterisation. Nonetheless, each of these results mean that it's a standard practise to attempt to write a stochastic process as an exponential process with respect to some continuous local martingale.

From prior courses, we've observed the lesser used maximal inequality for supermartingales that

$$\lambda\mathbb{P}\left(\sup_{s\le t}|X_s| \ge \lambda\right) \le \mathbb{E}\left[X_0\right] + 2\mathbb{E}\left[X_t^-\right].$$

Using the exponential process, we can improve this if $X \in \mathcal{H}^{2,c}$ and there exists a deterministic bound $\langle X\rangle_s \le a(s)$ for $s \le t$, to

$$\mathbb{P}\left(\sup_{s\le t} X_s \ge \lambda a(t)\right) \le \exp\left(-\frac{1}{2}\lambda^2 a(t)\right).$$

.

Stochastic Differential Equations

Theorem 9 (Girsanov's theorem) *Let (M_t) be a continuous local martingale on $(\Omega, \mathcal{F}, \mathcal{F}_t, \mathbb{P})$ up to time T. Then with*

$$Z_t := \frac{\mathrm{d}\mathbb{Q}|_{\mathcal{F}_t}}{\mathrm{d}\mathbb{P}|_{\mathcal{F}_t}},$$

the Radon-Nikodym derivative of the measures at time t (essentially, a random variable which returns how much more likely the sampled event is under $\mathbb{Q}$ than $\mathbb{P}$), we have that

$$X_t = M_t - \int_0^t \frac{\mathrm{d}\langle M, Z\rangle_s}{Z_s}$$

is a continuous local martingale on $(\Omega, \mathcal{F}, \mathcal{F}_t, \mathbb{Q})$ up to time T.

Note we require that $\mathbb{Q} \ll \mathbb{P}$ in order for this quantity to be defined, as otherwise Z_t is ill-defined on $\mathbb{P}^{-1}(0) \setminus \mathbb{Q}^{-1}(0)$.

This result may be seen readily via noting that

$$\mathbb{E}^{\mathbb{Q}}\left[\chi_A(X_t - X_s)\right] = \mathbb{E}^{\mathbb{P}}\left[\chi_A(Z_t X_t - Z_s X_s)\right]$$

and thus we can use integration by parts to get our result.

Equivalently, $M_t - \langle M, L\rangle_t$ is a continuous local martingale with respect to $\mathbb{Q}$, where (L_t) satisfies the exponential equation

$$\mathrm{d}Z_t = Z_t\,\mathrm{d}L_t$$

A corollary of this is that with L a continuous local martingale starting at 0, such that $\mathcal{E}(L)$ is a continuous martingale up to T, then with $\mathbb{Q}$ defined by

$$\frac{\mathrm{d}\mathbb{Q}|_{\mathcal{F}_t}}{\mathrm{d}\mathbb{P}|_{\mathcal{F}_t}} = \mathcal{E}(L)_t,$$

for any M a continuous local martingale in $\mathbb{P}$, $M - \langle M, L\rangle$ is a continuous local martingale in $\mathbb{Q}$.

The point of this theorem (**is this the point of this theorem?**) is to say that when we change our measure for one not so different (strictly absolutely continuous with respect to), we maintain the class of semimartingales. Every semimartingale with respect to $\mathbb{P}$ is also a semimartingale with respect to $\mathbb{Q}$, with an added drift.

Theorem 10 (Martingale representation theorem) *Let (M_t) be a square-integrable martingale on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$. Then there is a process $F = (F_t) \in \bigcap_{T\ge 0} L^2[0, T]$ such that*

$$M_t = \mathbb{E}\left[M_0\right] + \int_0^t F_s\,\mathrm{d}B_s$$

a.s. for any $t \ge 0$. In particular, any martingale with respect to the Brownian filtration $(\mathcal{F}_t)$ has a continuous version.

While this is a nice result (and presumably has implications on a wider level), for the purpose of this course it's got few applications, and mainly serves to demonstrate the generality of the exponential martingale process. As of writing (prior to the 2026 paper), no paper in the last 6 years has required it.

The proof idea is in a few steps. Firstly, note that we need only consider M_T for some fixed $T > 0$, as then using the regularity of F,

$$\begin{aligned} M_t &= \mathbb{E}\left[M_T \,|\, \mathcal{F}_t\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[M_T\right] + \int_0^T F_s\,\mathrm{d}B_s \,\middle|\, \mathcal{F}_t\right] \\ &= \mathbb{E}\left[M_0\right] + \int_0^t F_s\,\mathrm{d}B_s. \end{aligned}$$

We can then construct F by taking T arbitrarily large.

Now working with a square-integrable $\xi \in L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ (noting that we're considering L^2 over two different domains in this proof – one temporal, the other probabilistic), we want to show that

$$\xi = \mathbb{E}\left[\xi\right] + \int_0^T F_s\,\mathrm{d}B_s.$$

The idea here is to keep reducing the problem. It may be shown through functional analytic techniques that

$$\mathbb{L} = \mathrm{Span}\left\{\mathcal{E}\left(\int_0^\cdot h(s)\,\mathrm{d}B_s\right)_T : h \in L^2[0, T]\right\}$$

is dense in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$, and under these conditions it's then sufficient to show that (via the exponential property)

$$\mathcal{E}\left(\int_0^\cdot h(s)\,\mathrm{d}B_s\right)_T = 1 + \int_0^T \mathcal{E}\left(\int_0^\cdot h(s)\,\mathrm{d}B_s\right)_t h(t)\,\mathrm{d}B_t,$$

giving us our F.

Thus our proof rests on the above claim, that $\mathbb{L}$ is dense in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$. This takes a couple of steps: firstly, we show that

$$\{\varphi(B_{t_1}, \ldots, B_{t_k}) : k \ge 1, (t_i) \in [0, T], \varphi \in C_0^\infty(\mathbb{R}^k)\}$$

is dense in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$. This follows by constructing for an arbitrary $X \in L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ a martingale sequence $\mathbb{E}\left[X \,|\, B_{t_1}, \ldots, B_{t_n}\right]$ (by some enumeration of $\mathbb{Q} \cap [0, T]$), and then using the martingale convergence theorem to get convergence in L^2. Each of these approximating variables is then an L^2 function of $(B_{t_1}, \ldots, B_{t_n})$, and so we can use the density of $C_0^\infty(\mathbb{R}^n)$ in $L^2(\mathbb{R}^n, \mu)$ (for μ the Gaussian density on the Brownian motions, using that this is absolutely continuous with respect to leb).

We may then utilise this density to show the density of the exponentials, by using that for h a step function, the stochastic part of $\mathcal{E}(h \cdot B)_T$ is then analytic in finitely many B_{t_i}. We show that $\mathbb{E}\left[H\mathcal{E}(h \cdot B)_T\right] = 0$ iff $H = 0$ by continuing along these lines, and crucially by using the Fourier representation of analytic functions. This shows the result, as for U a Hilbert subspace, if the orthogonal component $U^\top = \{0\}$, then $\overline{U} = (U^\top)^\top = \{0\}^\top$ which is the entire space, and thus U is dense in the space.

SDEs

Definition 4 *An adapted continuous $\mathbb{R}^N$-valued stochastic process (X_t) on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ is a weak solution of*

$$\mathrm{d}X_t^j = \sum_{i=1}^n f_i^j(t, X_t)\,\mathrm{d}B_t^i + f_0^j(t, X_t)\,\mathrm{d}t \quad j \in \{1, \ldots, N\}$$

if there is a Brownian motion (B_t) in $\mathbb{R}^N$ adapted to $(\mathcal{F}_t)$ such that

$$X_t^j - X_0^j = \sum_{i=1}^n \int_0^t f_i^j(s, X_s)\,\mathrm{d}B_t^i + \int_0^t f_0^j(s, X_s)\,\mathrm{d}s.$$

It is a strong solution if we have the same, but $(\mathcal{F}_t)$ is the natural filtration of the Brownian motion.

Intuitively, a strong solution requires no more randomness to be added, whereas a weak solution may involve writing $X_t = f(t, (B_s)_{s\le t}, \xi_t)$ for ξ some 'additional' randomness.

With this in place, we say that a solution is *path-wise unique* if any two solutions on the same space with the same starting condition and Brownian motion must be equal, and it is *unique in law* if two solutions (with possibly different Brownian motions) started with the same initialisation distribution must have the same distribution. It is an intuitive result that path-wise uniqueness implies uniqueness in law (although not trivial to prove).

As for PDEs, it is possible to solve linear SDEs explicitly. Take

$$\mathrm{d}X_t^{(i)} = \sum_{j=1}^n \sigma_{ij}\mathrm{d}B_t^{(j)} + \sum_{k=1}^N \beta_{ik}X_t^{(k)}\mathrm{d}t$$

for $i \in [N]$. Writing this as $\mathrm{d}X_t = \sigma\mathrm{d}B_t + \beta X_t\mathrm{d}t$, we can have the intuition that this should look like $e^{\beta t}$ (in the sense that works for a matrix). We thus apply Itô's formula to $Y_t := e^{-\beta t}X_t$, and see that

$$X_t = e^{\beta t}X_0 + \int_0^t e^{\beta(t-s)}\sigma\mathrm{d}B_s$$

solves the SDE. As the integrand is deterministic, it is straightforward to see that X_t has mean $e^{\beta t}\mathbb{E}\left[X_0\right]$, and variance via Itô's isometry is $\int_0^t e^{2\beta(t-s)}\sigma^2\mathrm{d}s = \sigma^2(e^{2\beta t - 1})/2\beta$. This forms the notion of a diffusion process.

Definition 5 (Cameron-Martin's formula) *The Cameron-Martin's formula is*

$$\mathrm{d}X_t = \mathrm{d}B_t + b(t, X_t)\mathrm{d}t$$

for $b(t, x)$ a bounded, Borel measurable function.

This formula is important, as we are able to apply Girsanov's theorem to solve it. With W a standard BM, and

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_t} = \mathcal{E}\left(\int_0^\cdot b(s, W_s)\mathrm{d}W_s\right)_t,$$

as this is a martingale, thus $B_t := W_t - W_0 - \langle W, b(\cdot, W)\cdot W\rangle_t$ is a continuous local martingale in $\mathbb{Q}$ by Girsanov's theorem. Furthermore, the quadratic variation of B_t is t, and so by Levy's characterisation B_t is a $\mathbb{Q}$-Brownian motion. Furthermore,

$$\mathrm{d}B_t = \mathrm{d}W_t - b(t, W_t)\mathrm{d}t$$

so W_t solves our formula within $\mathbb{Q}$, providing a weak solution. This then leads to a straightforward (if slightly technical) generalisation to the multidimensional case. We may furthermore show uniqueness in law by arguing that for an arbitrary solution, we can perform a measure change to a Brownian motion, which will allow us to write the distribution of the solution in terms of the Wiener measure and b, ensuring that it is unique in law.

Existence and uniqueness

Lemma 11 (Gronwall) *If we have $g, h : [0, T] \to \mathbb{R}$ with h integrable, $\alpha \in \mathbb{R}$ such that for $t \in [0, T]$,*

$$0 \le g(t) \le h(t) + \alpha\int_0^t g(s)\mathrm{d}s,$$

then

$$g(t) \le h(t) + \alpha\int_0^t e^{\alpha(t-s)}h(s)\mathrm{d}s.$$

This is an example of a bootstrapping lemma, as we're able to go from a quantity growing with respect to itself to argue that it therefore cannot blow up outside of the domain. Critically, we get a bound independent of g, and so we just want to find h which is ideally polynomial to bound g by exponential growth.

With this in place, we may then argue
Lemma 12 *Let B be a standard BM in $\mathbb{R}$ on $(\Omega, \mathcal{F}_t, \mathcal{F}, \mathbb{P})$, (Z_t), $(\widetilde{Z}_t)$ be two continuous adapted processes, $f(t, x)$ Lipschitz in the spatial argument, then with the continuous local martingale M defined as*

$$M_t = \int_0^t f(s, Z_s) - f(s, \widetilde{Z}_s)\mathrm{d}B_s,$$

we have

$$\mathbb{E}\left[\sup_{s\le t}|M_s|^2\right] \le 4C^2\int_0^t \mathbb{E}\left[(Z_s - \widetilde{Z}_s)^2\right]\mathrm{d}s$$

for $t \ge 0$.

Furthermore, with N a finite variation process defined as

$$N_t = \int_0^t f(s, Z_s) - f(s, \widetilde{Z}_s)\mathrm{d}s,$$

then

$$\mathbb{E}\left[\sup_{s\le t}|N_s|^2\right] \le C^2 t\int_0^t \mathbb{E}\left[(Z_s - \widetilde{Z}_s)^2\right]\mathrm{d}s$$

These inequalities are then critical for proving uniqueness.

Theorem 13 *With $f : [0, \infty) \times \mathbb{R}^N \to \mathbb{R}^{N\times n}$, $g : [0, \infty) \times \mathbb{R}^N \to \mathbb{R}^N$ B an n-BM, define the SDE:*

$$\mathrm{d}X_t = f(t, X_t)\mathrm{d}B_t + g(t, X_t)\mathrm{d}t.$$

Then if each f_{ij} and g_i are Lipschitz in their spatial argument, and are also bounded by $C(1 + |x|)$, then for any L^2 starting condition, there is a unique strong solution.

To argue this, we perform a Picard-like iteration to approximate the solution.

Moving slightly further, under the same conditions it can be shown that the solution is a.s. uniformly continuous in its (deterministic) starting position.

Martingale problem

In some cases, we want to convert the problem of finding a weak solution of an SDE where for $i \in [d]$,

$$\mathrm{d}X_t^i = \sum_{k=1}^d \sigma_k^i(X_t)\mathrm{d}B_t^k + b^i(X_t)\mathrm{d}t,$$

into a problem of determining if a functional of X is a local martingale.

To do this, note that for any $f \in C^2(\mathbb{R}^d)$,

$$\begin{aligned} \mathrm{d}f(X_t) &= \sum_{i=1}^d \frac{\partial f}{\partial x^i}(X_t)\mathrm{d}X_t^i + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d \frac{\partial^2 f}{\partial x^i\partial x^j}\mathrm{d}\langle X^i, X^j\rangle_t \\ &= \sum_{i=1}^d \frac{\partial f}{\partial x^i}(X_t)\left(\sum_{k=1}^d \sigma_k^i(X_t)\mathrm{d}B_t^k + b^i(X_t)\mathrm{d}t\right) \\ &\quad + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d \frac{\partial^2 f}{\partial x^i\partial x^j}(X_t)\left(\sum_{k=1}^d \sigma_k^i(X_t)\sigma_k^j(X_t)\mathrm{d}t\right) \\ &= \sum_{i=1}^d\sum_{k=1}^d \frac{\partial f}{\partial x^i}(X_t)\sigma_k^i(X_t)\mathrm{d}B_t^k + Lf(X_t)\mathrm{d}t \end{aligned}$$

where

$$Lf(x) = \sum_{i=1}^d b^i(x)\frac{\partial f}{\partial x^i}(x) + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d \frac{\partial^2 f}{\partial x^i\partial x^j}(x)\sum_{k=1}^d \sigma_k^i(x)\sigma_k^j(x)$$

and thus defining $M_t^{[f]}$ to solve, with $M_0^{[f]} = 0$,

$$\mathrm{d}M_t^{[f]} = \mathrm{d}f(X_t) - Lf(X_t)\mathrm{d}t$$

we get

$$\mathrm{d}M_t^{[f]} = \sum_{i=1}^d \frac{\partial f}{\partial x^i}(X_t)\sum_{k=1}^d \sigma_k^i(X_t)\mathrm{d}B_t^k$$

and thus M_t is a continuous local martingale.

We then say that a process (X_t) on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ solves the L-martingale problem if for $f \in C^2(\mathbb{R}^d)$,

$$M_t^{[f]} = f(X_t) - f(X_0) - \int_0^t Lf(X_s)\mathrm{d}s$$

is a continuous local martingale. Indeed this is basically accurate – any weak solution to the SDE will solve the L-martingale problem provided σ and b are bounded and Borel measurable. The reverse direction is slightly more difficult, but essentially: if σ is uniformly positive definite with eigenvalues bounded above and below, then any solution to the L-martingale problem is a solution to the SDE.

An advantage of this formulation is that we don't require a definition of Brownian motion, and thus this approach can be generalised to manifolds.

Local times

We now seek to find cases where we can generalise the functions f for which Itô's formula applies. Indeed it is possible to show that if f is convex and X is a continuous semimartingale, then $f(X)$ is again a semimartingale with a similar formula, known as the generalised Itô's formula.

To show this is essentially a task of prelims and part A analysis using a mollifier $\alpha \in C_0^\infty(\mathbb{R})$ with $\int_0^\infty \alpha(s)\,\mathrm{d}s = 1$. Writing $f_\varepsilon = (x \mapsto |x - a|) * \alpha_\varepsilon$ where $\alpha_\varepsilon(s) = \frac{1}{\varepsilon}\alpha(s/\varepsilon)$, $f_\varepsilon \xrightarrow{u} (x \mapsto |x - a|)$ as $\varepsilon \to 0$, so one can then take limits in probability on Itô's formula to get an adapted continuous process for any continuous semimartingale X,

$$L_t^a = \frac{1}{2}\left(|X_t - a| - |X_0 - a| - \int_0^t \mathrm{sgn}(X_s - a)\,\mathrm{d}X_s\right).$$

This is a definition of the local time of X at a. The intuition here is that when (X_t) is away from a, $|X_t - a|$ changes linearly in X_t, and so the continuous martingale process $\mathrm{sgn}(X_t - a)\mathrm{d}X_t$ captures all of the change. $|X_t - a|$ is not quite a continuous local martingale however, due to its behaviour around a.

To provide a heuristic: the discontinuity of the derivative of $x \mapsto |x - a|$ means that where $X_t + \mathrm{d}X_t < a < X_t$ or $X_t \le a < X_t + \mathrm{d}X_t$, in each case $|X_t - a| + \mathrm{sgn}(X_t - a)\mathrm{d}X_t = -|X_t + \mathrm{d}X_t - a|$ (put into words: around a, the weak derivative $\mathrm{sgn}(X_t - a)$ pushes the process too far over a, requiring a correction). Thus we see that, writing $|X_t + \mathrm{d}X_t - a| = |X_t - a|$,

$$\mathrm{d}|X_t - a| - \mathrm{sgn}(X_t - a)\mathrm{d}X_t = \begin{cases} 2|X_t - a| & \text{if } \mathrm{sgn}(X_t - a) \ne \mathrm{sgn}(X_t + \mathrm{d}X_t - a) \\ 0 & \text{otherwise.} \end{cases}$$

and indeed this defines $2\mathrm{d}L_t^a$ almost exactly (and explains the constant 2)! This gives us a stochastic process L_t^a which accumulates by $\mathrm{d}L_t^a = |X_t - a| \in [0, |\mathrm{d}X_t|]$ when X_t is close to a, and remains constant elsewhere. As we will see, there is a strong sense in which L_t tracks the quadratic variation of X_t around a (or more roughly, the space covered by the 'jitters' of the process over a).

In particular, $\int_0^t |X_s - a|\mathrm{d}L_s^a = 0$, which may be seen through IBP.

We claim that if X is a continuous semimartingale, then for *any* convex function f, $f(X_t)$ is a continuous semimartingale. This argument involves writing

$$f(x) = \alpha_n + \beta_n x + \frac{1}{2}\int_{-n}^n |x - z|\mathrm{d}f'_+(z)$$

and then applying Tanaka's formula for L_t^a localised by hitting times of n, then taking $n \to \infty$, we achieve

$$f(X_t) = f(X_0) + \int_0^t f'_-(X_s)\mathrm{d}X_s + \int_{-\infty}^\infty L_t^z\mathrm{d}f'_+(z).$$

Indeed if no discontinuity exists and $f \in C^2$, we may apply Itô's standard formula to observe that

$$\frac{1}{2}\int_0^t f''(X_s)\mathrm{d}\langle X\rangle_s = \int_{-\infty}^\infty L_t^z f''(z)\mathrm{d}z.$$

giving a strong sense in which $2L_T^z\mathrm{d}z = \mathbf{1}_{t\le T}\mathrm{d}\langle X\rangle_t$ (for continuous non-negative, and thus also bounded measurable functions). We can then go further with the below result.

Theorem 14 (Occupation time) *If X is a continuous semimartingale, and $\int_0^t \mathbf{1}_{\{a\}}(X_s)\mathrm{d}A_s \equiv 0$, then*

$$\lim_{\varepsilon\to 0}\frac{1}{4\varepsilon}\int_0^t \mathbf{1}_{(a-\varepsilon, a+\varepsilon)}(X_s)\mathrm{d}\langle X\rangle_s = L_t^a.$$

In particular, note that this applies for a standard BM.

This is immediate from the above. We may then apply this to prove a comparison theorem.

Theorem 15 (Ordering of SDEs) *Let σ, $b^{(1)}$ and $b^{(2)}$ be real-valued Lipschitz continuous functions on $\mathbb{R}$, and let $X^{(1)}$ and $X^{(2)}$ be strong solutions to*

$$\mathrm{d}X_t^{(i)} = \sigma(X_t^{(i)})\mathrm{d}B_t + b^{(i)}(X_t^{(i)})\mathrm{d}t.$$

Then if $X_0^{(1)} \ge X_0^{(2)}$ almost surely, and $b^{(1)} \ge b^{(2)}$, then $X_t^{(1)} \ge X_t^{(2)}$ almost surely.

This result uses Tanaka's formula to evaluate the negative part of the difference between the two, and identify that it is identically zero almost surely.

Theorem 16 (Regularity of local time) *Let $X = X_0 + M + A$ be a continuous semimartingale, L_t^a be its local time at a. Then almost surely, L_t^a is jointly continuous in $t \in (0, \infty)$ and right-continuous in $a \in \mathbb{R}$ (so any convergent sequence (t_n, a_n) where (a_n) is decreasing will have $L_{t_n}^{a_n}$ converge). Furthermore, L_t^{a-} exists almost surely, and where it does*

$$L_t^a - L_t^{a-} = \int_0^t \chi_{X_s = a}\mathrm{d}A_s.$$

Thus we get full joint continuity a.s. wherever this integral is zero.

Note firstly that, provided we localise to bound M, A, and $V(A)$, the only problematic (not trivially continuous) term in L_t^a is the $\int_0^t \mathrm{sgn}(X_s - a)\mathrm{d}M_s$ for M the martingale part of X (continuity and finite variation allow us to deal with the other side). For this term, we use the Kolmogorov-Čentsov criterion, giving Hölder continuity of stochastic process sample paths when we have a stronger continuity of the moments of their differences:

$$\begin{aligned} \mathbb{E}\left[|N(a)_t - N(b)_s|^{2p}\right] &= \mathbb{E}\left[\left|\int_0^t \mathrm{sgn}(X_u - a)\mathrm{d}M_u - \int_0^s \mathrm{sgn}(X_u - b)\mathrm{d}M_u\right|^{2p}\right] \\ &= \mathbb{E}\left[\left|\int_s^t \mathrm{sgn}(X_u - a)\mathrm{d}M_u + \int_0^s \chi_{(b,a]}(X_u)\mathrm{d}M_u\right|^{2p}\right] \\ &\le 2^{2p-1}\left(\mathbb{E}\left[\left|\int_s^t \mathrm{sgn}(X_u - a)\mathrm{d}M_u\right|^{2p}\right] + \mathbb{E}\left[\left|\int_0^s \chi_{(b,a]}(X_u)\mathrm{d}M_u\right|^{2p}\right]\right) \end{aligned}$$

where we use $|(x + y)/2|^p \le (|x|^p + |y|^p)/2$, plus the fact that the two martingales inside this sum are orthogonal. From here we may progress using BDG, plus being able to bound the latter term by $|a - b|^p$ using a separate argument with the functions $x \mapsto \frac{1}{2}(x - a)^2\chi_{x\le a}$. Subject to localisation, these all allow us to get continuity. To extend this then to the general case is done using the time-change $\tau_t = \inf\{s : \langle M\rangle_s + V(A)_s + s > t\}$.

Theorem 17 (Regularity of Brownian motion local time) *With L_t^a the local time of a one-dimensional standard Brownian motion W, for $L, T > 0$, $p > 1$ there is a constant C such that for $t, s \in [0, T]$, $a, b \in [-L, L]$,*

$$\mathbb{E}\left[|L_t^a - L_s^b|^{2p}\right] \le C(|b - a|^p + |t - s|^p).$$

Furthermore, L_t^a is jointly Hölder continuous for $\alpha < 1/2$.

This comes out as an improvement of the prior proof.

Theorem 18 (Skorokhod's equation) *If $y \in C[0, \infty)$ is a continuous path in $\mathbb{R}$ with $y(0) \ge 0$, $k(t) = \left(\inf_{0\le s\le t} y(s)\right)^-$, then k is the unique continuous non-decreasing function on $[0, \infty)$:*

- *beginning at 0;*
- *such that $y + k \ge 0$;*
- *and which increases only on $(y + k)^{-1}(\{0\})$, so $\int_0^\infty (1 - \mathbf{1}_{\{0\}}(y(t) + k(t)))\,\mathrm{d}k(t) = 0$.*

This proof is just a matter of deterministic analysis.

The consequence for our purposes comes from noting the relation to Tanaka's formula. $2L_t^a$ in this case reflects exactly the definition of k in relation to

$$y(t) = \int_0^t \mathrm{sgn}(X_s - a)\,\mathrm{d}X_s + |X_0 - a|.$$

This is because we can show that $\int_0^t |X_s - a|\mathrm{d}L_s^a = 0$ using Itô, Tanaka, and IBP, and consequently that L_t^a increases only on $X_s = a$. This gives us that

$$L_t^a = \frac{1}{2}\left(|X_0 - a| + \inf_{0\le s\le t}\int_0^s \mathrm{sgn}(X_r - a)\,\mathrm{d}X_r\right)^-.$$

LaTeX TikZposter